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which are included within the family. Variation occurs in number, arrangement, adhesion and cohesion of the floral parts. We study a few families for understanding the process and purpose of classification.

5.13.1 Family: Fabaceae (Pea family)

Systematic position

APG classification		Bentham and Hooker classification	
Kingdom	Plantae	Kingdom	Plantae
Clade	Angiosperms	Class	Dicotyledonae
Clade	Eudicots	Sub-class	Polypetalae
Clade	Rosids	Series	Calyciflorae
Order	Fabales	Order	Rosales
Family	Fabaceae	Family	Fabaceae



Diagnostic features

- Leaves simple or imparipinnately compound or palmate, leaf base pulvinate, leaflets stipellate.
- Flowers Zygomorphic
- Corolla: Papilionaceous, descendingly imbricate aestivation, posterior petal outermost,
- Petals clawed.
- Stamens: Monadelphous, diadelphous
- Ovary stipitate (a short stalk as the base), monocarpellary, unilocular with marginal placentation.
- Fruit a legume or lomentum.

General characters

Distribution: Fabaceae includes about 741 genera and more than 20,200 species. The members are cosmopolitan in distribution but abundant in tropical and subtropical regions.

Habit: All types of habits are represented in this family. Mostly herbs (*Indigofera, Crotalaria*), prostrate (*Indigofera enneaphylla)* erect (*Crotalaria verrucosa*), shrubs (*Cajanus cajan*), small trees (*Sesbania*), climbers (*Clitoria*), large tree (*Pongamia, Dalbergia, Erythrina, Butea*), woody climber (*Mucuna*), hydrophyte (*Aeschynomene aspera*) commonly called **pith plant.**



Root nodule

Root: Tap root system, roots are nodulated, have tubercles containing nitrogen – fixing bacteria (*Rhizobium leguminosarum)*

Stem: Aerial, herbaceous, woody (*Dalbergia*) twining or climbing *Clitoria*.

Leaf: Leaf simple or unifoliate (*Desmodium gangeticum*) bifoliate (*Zornia diphylla,*), Trifoliate (*Lablab purpureus)*, unipinnate or simple pinnate (*Clitoria*), alternate, stipulate, leaf base, **pulvinate**, stipulus 2, free. Leaves showing reticulate venation terminal leaflet modifies into a **tendril** in *Pisum sativum.*

Inflorescence: Raceme (*Crotalaria verrucosa*), panicle (*Dalbergia latifolia*) axillary solitary (*Clitoria ternatea*)